appear dark violet.

Step 3: Next, the slide is decolurized with alcohol or an acetone alcohol solution. This solution is a decolurizing agent, which removes the primary stain from the cells of some species but not from others.

Step 4: The slide is immediately washed after decolurization and the slide is then counter stained with basic fuchsin or safranin, a basic red dye. The smear is washed again, blot dried and examined under microscope (Figure 3.7).

3.7.3 Principle of Gram's Staining

The exact mechanism of action of this staining technique is not clearly understood. However, the most acceptable explanations are associated with the structure and composition of the cell wall.

The cell wall of Gram positive bacteria have a thicker peptidoglycan (consists of disaccharides and amino acids) than Gram negative bacteria. Figure 3.8 depicts the cell wall of Gram positive and Gram negative bacteria. In addition, Gram negative bacteria contain a layer of lipo polysaccharide (consists of lipids and polysaccharide) as part of their cell wall. When Crystal violet and subsequently Iodine is applied to both Gram positive and Gram negative cells, the two combine to form CV-I complex.

The cell wall of Gram positive bacteria with lower lipid content get dehydrated during alcohol treatment. The pore size decreases and the permeability is reduced. Thus, the CV-I complex cannot be extracted and the cells remain violet.

The alcohol treatment of Gram negative bacteria extracts the lipid which results in increased porosity or permeability of the cell wall. Thus, the crystal violet iodine [CV-I] complex is extracted and the bacteria are decolorized. These cells subsequently take on the colour of the counter stain basic fuchsin or safranin and appears red to pink.

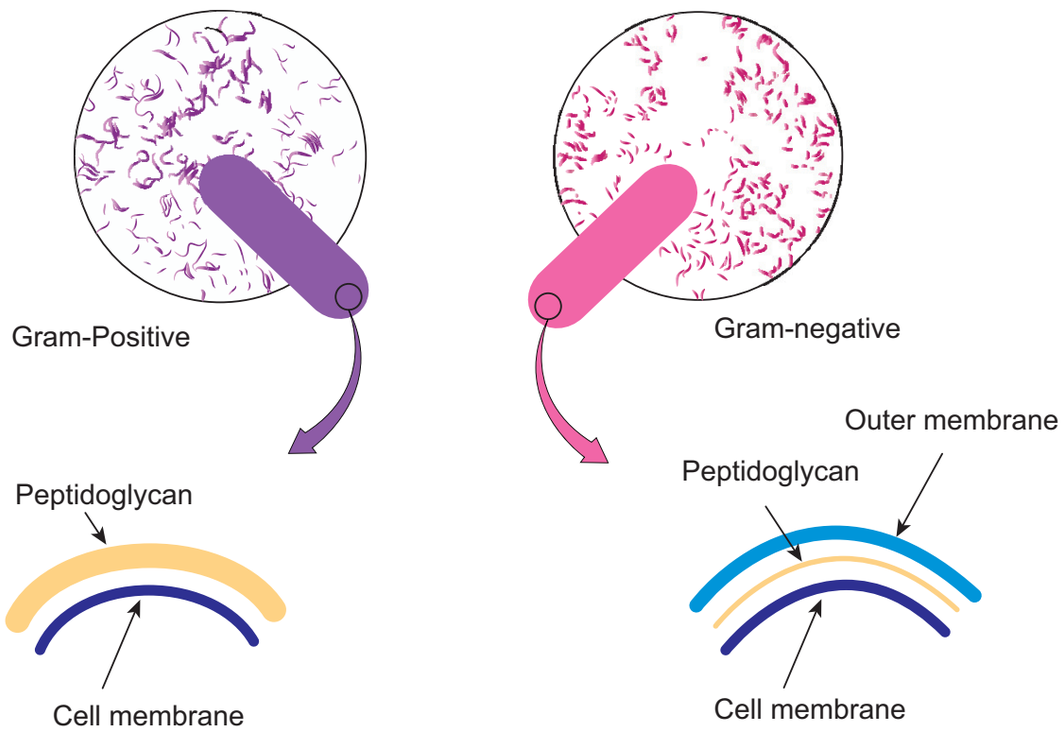


Figure 3.8: Cell wall of Gram positive and Gram negative Bacteria

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